INCUBRIX ENGINEERING ASSESSMENT 4

Distribution: Long-Tail Destination Network

Take IncuBrix publishing beyond the big platforms - into the communities, publications and open networks where real audiences gather

Thank you for applying to IncuBrix

IncuBrix publishes creators' content to many destinations in one action. In this assessment you will build the next stage of our Distribution product. Please read the whole brief once before you start.

At a glance	
What you build	Publishing connectors and the intelligence around them: connectors for forums, open networks, developer blogs, newsletters and partner sites; automatic rule checks so we never spam; proof that every destination works; and per-destination tailoring.
Why it matters	The big platforms are already connected. Growth now comes from hundreds of smaller, highly relevant destinations - but only if posting there is reliable, rule-respecting and proven.
What you hand over	A containerised connector service built to our connector contract, a live demo URL, your code, a short report and a test pack.
Time allowed	10 calendar days from the day you receive this brief. The exact deadline is in the email.
Tools allowed	Anything: any AI coding assistant, model, API, cloud or library. Just disclose it and keep it commercially usable.
How you are selected	You fill in a Submission Form showing what works, with proof. We verify it and run 5 hidden tests (screening, 80 marks), then hold a 60-90 minute live session (20 marks) with the strongest candidates. Shortlist: 75+. Top pick: 85+.

How to use this brief

1. Read **The problem** and **What you build** to understand the job.
2. Build the **Must-haves** first. Each one is pass or fail, and every one must pass.
3. Then climb the **Levels**: Core is required; Advanced and Outstanding earn more marks.
4. Follow the **Integration Standard** so your work can plug straight into our product.
5. Fill in the **Submission Form** (Excel file sent with this brief) as you go. It is how you show us your work, and it is the main thing we review.

The problem

IncuBrix Distribution already publishes to the 15 major platforms (YouTube, Instagram, Facebook, LinkedIn, TikTok, X, Threads, Pinterest, Bluesky, Mastodon, Reddit, Vimeo, WordPress, Telegram and Discord) and to RSS feeds. These are **out of scope**; do not rebuild them.

The next level is the long tail: Discourse forums, Lemmy communities, developer publishing (Dev.to, Hashnode), blogs (Ghost), newsletters, Slack communities, partner websites, video feeds for CTV and video aggregators, and communities where automated posting is not allowed and a person must post. There are hundreds of these, each with its own format limits and its own rules. One careless post can get IncuBrix banned from a community.

The key idea: one connector, many destinations

A destination is one place that reaches one audience: one forum, one community, one newsletter. A connector is the code for one kind of platform. One Discourse connector must serve every Discourse forum, so adding forum number 41 should be a single configuration row, not new code. And no destination counts until it is **certified**: its format limits and posting rules are recorded and a real test post has succeeded and been cleaned up.

Out of scope

The 15 major platforms and RSS (already live); paid boosting and ads; lead generation; cold email and outreach; billing.

What you build

1. **Connectors** that follow the IncuBrix connector contract exactly (Appendix A): the same methods, manifest format and error codes as our existing connectors, so our runtime can load them unchanged.
2. **Posting-rules engine:** each destination's rules (how often we may post, whether self-promotion is allowed, link rules, AI-disclosure rules) are stored and checked before every post.
3. **Certification harness:** for each destination, publish a test post, confirm it is live, read its status and delete it (or post to an agreed sandbox), and store the evidence.
4. **Destination catalogue tools** (Advanced): import a spreadsheet of destinations, clean it, score each one for relevance to IncuBrix's audience, and capture its rules.
5. **Tailoring and recommendation** (Advanced and Outstanding): adapt the title, text and format for each destination, and recommend where a piece of content should go.

IncuBrix's audience for relevance scoring: AI, video, marketing, SaaS, creators and founders.

Must-haves (all must pass)

#	Must-have	How we verify it (you give the evidence in your form)
1	**Exact connector contract.** Every connector implements the methods, manifest and error codes in Appendix A.	Our contract test suite runs against your service.
2	**Six Core connectors certified:** Discourse, Lemmy, Dev.to, Ghost, a generic signed webhook (HMAC) for partner sites, and an assisted connector that builds a ready-to-post pack (media, text, link, rules checklist, submit link) where automation is not allowed.	Certification evidence plus our own runs against the local test instances.
3	**Configuration, not code.** Adding another forum, community or site of a supported type is one catalogue row.	We add a new instance during review.
4	**Rules checked before every post:** frequency cap, self-promotion policy, link policy and AI-content disclosure. A post that breaks a rule is blocked or rescheduled with a reason.	We send a second post inside a weekly cap and a promotional post to a no-promotion destination.
5	**Never spam, never cheat.** Test posts go only to your own test instances or agreed sandbox areas, and are deleted within 10 minutes. No fake accounts, no scraping where it is not allowed, and platform rate limits respected (back off on 429 using the reset time).	We review your logs, code and evidence.
6	**Safe retries.** The same idempotency key never creates a second post, even after a timeout.	We force a timeout and retry.
7	**Secrets protected and simulation mode.** Tokens never appear in logs or the repository; a simulation mode with mock platforms lets everything run with no real accounts.	We search logs and the repository; we run in simulation mode.
8	**Local test instances.** `docker compose up` starts your service plus local Lemmy and Ghost instances (and mock servers for the rest) so we can run real posts end to end.	We run your compose file on a clean machine.

#	Must-have	How we verify it (you give the evidence in your form)
9	**Integration Standard** followed and the **live demo** works for the whole review period.	See the Integration Standard section.

Levels

Level	What you need to show
Core (required)	• All must-haves pass. • Each Core connector passes the full certification run (publish, confirm live URL, status, delete) with stored evidence. • Errors from each platform are mapped to the shared error codes, including expired tokens, rate limits and rejected content.
Advanced	• **Catalogue import:** a CSV of destinations is validated, de-duplicated (normalised URL plus platform), checked for reachability and enriched (for example member count); 1,000 rows import in under 5 minutes with a row-by-row error report. • **Rules capture:** an AI step reads a community's rules page and proposes its posting rules; a person confirms them before they are used. • **Relevance score** from 1 to 5 against IncuBrix's audience, with a short explanation for each destination. • **Tailoring:** each destination gets a fitting title and text, for example discussion-first framing for forums, a canonical URL for articles, truncation at a word boundary, correct hashtag and mention style. • **Three more connectors** from: Hashnode, Slack app, Kit or Mailchimp newsletter, MRSS video feed, PeerTube.
Outstanding	• **Recommender:** given a content item, rank destinations by fit (audience, rules, past results) with a reason for each, and measure precision of the top 10 on a set you label. • **Metrics:** fetch_metrics works on at least three connectors, normalised to views, likes, comments, shares and clicks. • **Recertification:** a scheduler re-tests destinations before 30 days; a failing destination is demoted and excluded automatically, with a ticket or alert. • **Real catalogue:** at least 150 real, relevant destinations researched with rules, relevance score and connector type (catalogue only - no posting to them).

Hard cases we will test

Build and test your system against situations like these, and show the results in the Hard cases tab of your Submission Form. During screening we run **5 hidden tests** of the same kinds on your live demo. A good system handles each one sensibly and says what it did (in warnings or error messages) instead of failing silently.

- A publish call that times out and is retried with the same idempotency key.
- A 429 response with a reset time; an expired token; content rejected by the platform.
- A title or body longer than the destination allows; media the destination does not accept.
- A second post inside a destination's frequency cap; a promotional post to a community that bans self-promotion.
- A platform that renames a field in its API: certification must fail and the destination must be demoted.
- A CSV with duplicates, bad URLs and mixed languages; a partner who revokes permission.
- A non-English community; a destination that requires AI-content disclosure.

How it must plug into IncuBrix (Integration Standard)

IncuBrix runs its services as containers that talk to each other over HTTP APIs. If your submission follows the ten points below, we can connect it to the product without rewriting it. Each point is worth 2 marks (20 marks in total). Tick each one in your Submission Form with a link to the proof.

Requirement	What exactly we check
1. One-command start and test	`docker compose up` starts the service and everything it needs (database, storage, local test servers). One command runs all tests (unit, integration and at least one end-to-end) with no API keys. A `.env.example` lists every setting.
2. Documented HTTP API	An `openapi.yaml` (OpenAPI 3.1) that matches the running service exactly. All routes under `/api/v1/`.
3. Background jobs	Anything slower than 2 seconds runs as a job: `POST` returns `202` with a `job_id`; `GET /api/v1/jobs/{job_id}` returns status (queued, running, succeeded, failed), progress 0-100 and result links. Optional callback URL, signed with HMAC-SHA256 in an `X-IncuBrix-Signature` header.
4. Safe retries	An `Idempotency-Key` header on every POST: sending the same request twice with the same key returns the same job and never does the work twice.
5. Clear errors	Errors are JSON: `{"error": {"code", "message", "retryable", "details"}}`. Codes are UPPER_SNAKE_CASE and listed in the README.
6. Configuration	All settings, paths, model names and limits come from environment variables or a config file. No hard-coded paths or keys.
7. Files in, files out	Inputs accepted as an upload or a URL. Outputs saved to S3-compatible storage (MinIO in the compose file is fine) and returned as download links.
8. Logs and health	Structured JSON logs with timestamp, level, job_id and the `X-Correlation-Id` from the request. `GET /health` and `GET /ready` endpoints.
9. Output manifest	Every result includes a JSON manifest: which models/providers and versions were used, settings, seed, timings, cost estimate, warnings, and licence/rights information.
10. Batch runner	A command-line tool that runs a folder or file of test inputs against the API and writes all results to one folder. We use this to test every submission the same way.

Tools and resources: use anything

We want to see the best system you can build, the way you would build it at work. There are no restrictions on tools.

- **AI coding assistants** (ChatGPT, Claude, Copilot, Cursor and so on): allowed, as much as you like.
- **Any AI model or service**: open-source or proprietary, self-hosted or hosted API, free or paid.
- **Any compute**: your laptop, a cloud GPU, a free notebook, a paid server.
- **Any open-source code or library**, as long as you credit it and its licence allows commercial use.

Because your solution should be able to go straight into the IncuBrix product, five rules apply:

1. **Disclose everything.** List every model, API, library, dataset and media source in SOURCES.md, and every AI coding tool you used (and for what) in AI_USE.md.
2. **Commercially usable.** Everything in your default path, and everything it outputs, must be licensed for commercial use. No research-only or non-commercial models, data or media (for example CC BY-NC) in the default path. Avoid AGPL code in the shipped service.
3. **Swappable providers.** Put every external AI service behind a small adapter selected by configuration, and include a mock adapter so tests run with no API keys.
4. **Know your cost.** Report the cost and time per unit of output (for example per minute of video). Paid services are fine, but you will not get extra marks for spending more. IncuBrix does not reimburse costs unless agreed in writing before you spend.
5. **No secrets in the code.** API keys live in environment variables or a secret store, never in the repository, logs or screenshots.

How we evaluate (simple, three steps)

You do most of the work of showing what you built: you fill in the **Submission Form** with what works and a link to the proof for each claim. We then check your claims and test your system ourselves.

Step	Who	What happens
1. Submission	You	Send the links and the completed Submission Form: must-haves met, levels reached, Integration Standard points done, numbers measured, hard cases tested. Every claim needs a link to evidence (a file, log, video timestamp, test or URL).
2. Screening	Us, about 45 minutes	We check that the must-haves really pass, verify your claims using your evidence and your live demo, and run 5 hidden tests. This gives a screening score out of 80.
3. Live session	You and us, 60-90 minutes	Only for candidates whose must-haves all pass and who score at least 60 of 80 in screening. Scored out of 20.

Marks (100)

Part	Marks	How the marks are given
Must-haves	Pass / fail	Every must-have must pass. One failure means the submission is not selected.
Levels reached	40	Core 15 + Advanced 15 + Outstanding 10. A level counts only when every item in it is verified, and only if the level before it is verified.
Integration Standard	20	2 marks for each of the 10 points we confirm.
Hidden tests	20	5 tests on your live demo, 4 marks each: 4 = handled correctly, 2 = handled with a clear flaw, 0 = wrong, crash or silent failure.
Live session	20	4 parts, 5 marks each (see the live session section).

Be precise and honest in your form

A claim without evidence scores 0. A claim that turns out to be false disqualifies the whole submission. It is far better to mark something "No" or "Partial" and explain why.

Numbers you must report (Metrics tab of the form)

Metric	How to measure it
Certification	Number of destinations certified per connector, time per certification run, and evidence links.
Reliability	Publish success rate in simulation and on local instances; duplicate posts under retry tests (must be 0).
Rule safety	Rule violations caught versus missed on our rule test set (any missed violation is serious).
Catalogue (Advanced)	Rows imported, duplicates and errors found, time for 1,000 rows.
Recommendation (Outstanding)	Precision of the top 10 on your labelled set.
Cost	Any API or model cost per 1,000 publications.

Who we select

We are hiring only the strongest engineers, so the bar is high and the same for everyone.

Result	What it takes
Top pick - hire	All must-haves pass, total of 85 or more, Outstanding level verified, and at least 14 of 20 in the live session.
Shortlist - hire	All must-haves pass, total of 75 or more, and at least 14 of 20 in the live session.
Not selected	Any must-have fails, screening below 60 of 80, total below 75, or live session below 14.
Disqualified	Any false claim in your form, fake or edited evidence, hidden manual work presented as automated, using someone else's work without credit, using people's voice, face or content without rights, or not being able to explain your own code.

If two candidates tie, the higher live-session score wins, then hidden tests, then Integration Standard.

The live session (60-90 minutes, online)

If all your must-haves pass and you score at least 60 of 80 in screening, we invite you to a live session. You share your screen and work in your own repository and environment. You may use documentation and your usual AI coding tools. Each part is scored 5 (excellent), 3 (good), 1 (weak) or 0 (not shown).

Part	What happens	What a 5 looks like
1. Walkthrough	You explain the architecture and trace one output field from the request through to the final file.	Clear, accurate, you know why each piece exists and what would break if it changed.
2. Predict and run	We give you an input you have not seen. You predict what your system will do, then run it.	Prediction matches; when it does not, you explain why within minutes.
3. Debugging	We break something (a setting, a dependency, a provider response) and you find and fix it.	You find the cause from logs and code quickly and fix it properly, not with a workaround.
4. Live change	For example: build a new connector for a platform we name, from your template, and certify it on a mock server; or add a new posting rule with a test.	You write or update a test first or alongside, make the change, run the tests and the end-to-end path, and it works.

What to send us

Reply to the invitation email before the deadline, attaching your completed Submission Form. Put all links in the form. Keep every link working until we announce results.

Item	What it must contain
Submission Form	The Excel file sent with this brief, fully completed: every must-have, level item and Integration Standard point marked Yes / Partial / No with a link to the evidence, plus your measured numbers and hard-case results. This is the main thing we review.
Live demo URL	A working URL plus an access token if you protect it. Tell us the cold-start time if there is one. Put a rate limit on it so it cannot be abused.
Repository	Private GitHub repository with reviewer access (the username is in the email). Real commit history, not one final commit.
README.md	What it does, architecture diagram, setup in under 10 commands, how to run tests and the batch runner, all settings.
openapi.yaml	The API contract (Integration Standard point 2).
REPORT.md	Two to four pages: the approaches you considered and why you chose yours, what works, what does not and why, measured numbers, cost per unit, and what you would do next.
SOURCES.md and AI_USE.md	As described in the tools section.

Item	What it must contain
Test pack	All inputs you tested with, all outputs (including failures), logs, and an index linking each claim in your report to a file.
Demo video	Unedited screen recording, at most 8 minutes: one clean run, one failure and how the system handles it, the tests running, and one design decision explained in code.
Certification pack	For every certified destination: the run record, live URL or API response, screenshot, and the deletion confirmation.
Catalogue files	Your catalogue CSV(s), import reports and, for Outstanding, the labelled set used to measure the recommender.

Rules, rights and using your work

- **Individual work.** One candidate, one submission. You may use any tools and open-source code, but the design and the core logic must be yours and you must be able to explain all of it.
- **Rights.** Use only media, audio, data and content that you own, that is public domain, or that is licensed for commercial use. Never use a real person's voice, face or content without their consent.
- **Label AI output.** Mark generated media as AI-generated in the manifest.
- **Honesty.** Keep failed attempts in your test pack. An honest report of what does not work scores better than a polished demo that hides problems.
- **Using your work in our product.** This assessment is designed so that a strong submission can go straight into IncuBrix. Your code remains yours. If we want to use it, we will agree the terms with you in writing (licence or ownership, and compensation) before any use. By submitting, you confirm that the work is yours and that it and everything it depends on can legally be used commercially.
- **Questions.** Reply to the invitation email. We answer within one working day and share useful answers with all candidates on the same assessment, so nobody gets an unfair advantage.

Quick answers

Question	Answer
Can I use ChatGPT, Claude, Copilot or similar?	Yes. Tell us how in AI_USE.md. In the live session you must be able to explain and change the code yourself.
Can I use paid APIs or a paid GPU?	Yes. Report the cost per unit. We do not reimburse costs unless agreed in writing first.
Can I start from an open-source project?	Yes, with credit and a licence that allows commercial use. We score what you added and how well it works.
Do I need a nice user interface?	No. A simple web page or the API docs page is enough. We score the engine, the integration and the results.
What if I cannot finish everything?	Make the must-haves and the Core level solid first. Then go as far up the levels as you can. Report honestly what is missing.
Do I need real accounts on every platform?	No. Use local instances (Lemmy, Ghost), free developer accounts where available, and mock servers for the rest. Never post to a real community without the owner's permission.

We look forward to seeing what you build. Good luck.

IncuBrix Engineering Team

Appendix A: the IncuBrix connector contract

Write connectors in Python 3.11+ or TypeScript (Node 20+). Expose each method over HTTP at `POST /connector/v1/{connector_key}/{method}` with JSON bodies, so our Connector Runtime can call them whatever language you choose.

Methods

```
validate(rendition, destination, connection)            -> {ok, violations[]}
publish(rendition, destination, connection, idempotency_key)
                                                        -> {platform_post_id, live_url, status}
get_status(platform_post_id, connection)                -> {status, live_url}
delete(platform_post_id, connection)                    -> {deleted}
fetch_metrics(platform_post_id, connection, since)      -> {views, likes, comments, shares,
                                                            saves, watch_time_s, clicks}   # optional
refresh_auth(connection)                                -> {expires_at}                    # optional
build_assist_pack(rendition, destination)               -> {files[], caption, deep_link,
                                                            checklist[]}          # assisted tier only
```

Methods return results only. They never write to IncuBrix's database; our runtime does that.

Manifest (one YAML file per connector)

```
key: discourse
version: 1.0.0
tier: native            # native | protocol | bridge | feed | assisted
auth: api_key           # oauth2 | api_key | bot_token | none
surfaces:
  - key: topic
    content_types: [text, image, video_link]
    spec: {title_max_chars: 255, body_max_chars: 32000, links_allowed: true}
rate_limits:
  per_connection: {requests: 60, per_seconds: 60}
capabilities: [publish, status, delete, metrics]
error_map:
  "429": PLATFORM_RATE_LIMIT
  "422": PLATFORM_CONTENT_REJECTED
  "401": CONNECTION_AUTH_EXPIRED
instance_param: base_url   # one connector, many destinations (one per forum)
```

Shared error codes

Code	Retry?	Meaning and expected action
DEST_SPEC_VIOLATION	No	Content breaks the destination's format limits; return field-level reasons.
DEST_RULE_FREQUENCY_CAP	Later	Frequency cap reached; reschedule after the cap window.
CONNECTION_AUTH_EXPIRED	After reconnect	Token expired or revoked; pause and ask the owner to reconnect.
PLATFORM_RATE_LIMIT	Yes	Back off using the platform's reset time.
PLATFORM_QUOTA_EXHAUSTED	Next window	Daily quota used up; reschedule.
PLATFORM_CONTENT_REJECTED	No	Platform refused the content; return its message for review.
PLATFORM_UNAVAILABLE	Yes	Retry; stop calling after 5 consecutive failures.
CONNECTOR_DISABLED	No	Kill switch is on for this connector.

Destination record your catalogue tools must produce

```
name, url, platform_key, surface, category, audience_tags, language, geo, est_audience,
rules_url, frequency_cap, self_promo_allowed, link_policy, disclosure_required,
automation_allowed, connector_key, relevance_score (1-5), relevance_reason, level, last_certified_at
```

Levels: L0 catalogued only; L1 assisted (a person posts); L2 bridged or feed; L3 native API (publish, status and delete tested); L4 native plus metrics.